

Geodesy: Ellipsoids, Datums, Projections, and Coordinate Systems

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Archaeological Research Design Decision!

What kind of coordinate system ?

- Local: It's easy to set up but HARD to relate your data to other spatial datasets.
- Global: It's harder, but getting easier, thanks to GPS. ***A global system is required the moment you want to combine your data with maps, satellite.***

Global Positioning Systems

- Personal
- Mapping grade
- Survey grade

DAACS Project #:		Department of Archaeology Context Record																							
130	Project Name: Site 30		Quadrat Number: 021	Context: B	Feature: /																				
Short Title: Dark reddish brown silty clay across quad			Coordinates: X <u>11498200</u> to <u>11498205</u> Y <u>3890710</u> to <u>3890715</u>																						
			Excavator: AC, KCD	Date Opened: 8/25/22	Recorder: AC																				
			Date Closed: 8/26/22																						
Elevations:			Stratigraphic Relationships:																						
<table border="1"> <thead> <tr> <th></th> <th>Opening</th> <th>Closing</th> <th>Opening - Closing:</th> </tr> </thead> <tbody> <tr> <td>NW</td> <td>640.96</td> <td>640.80</td> <td>0.16</td> </tr> <tr> <td>NE</td> <td>640.48</td> <td>640.30</td> <td>0.18</td> </tr> <tr> <td>SE</td> <td>640.47</td> <td>640.30</td> <td>0.17</td> </tr> <tr> <td>SW</td> <td>640.88</td> <td>640.70</td> <td>0.18</td> </tr> </tbody> </table>				Opening	Closing	Opening - Closing:	NW	640.96	640.80	0.16	NE	640.48	640.30	0.18	SE	640.47	640.30	0.17	SW	640.88	640.70	0.18	Sealed by: <u>A</u> Seals: <u>C</u> Intruded by: _____ Intrudes: _____ Correlates: <u>004B, 010B</u> Contains: _____ Within: _____ Other: _____		
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GPS and GNSS — and How Accurate Is “Accurate”?

- **GPS** is the U.S. system.
- **GNSS** (Global Navigation Satellite Systems) is the umbrella term for all of them: GPS (USA), GLONASS (Russia), Galileo (EU), BeiDou (China).
- Most modern receivers track several at once — more satellites means great accuracy

Tier	Typical accuracy	Example hardware	Good enough for...
Personal / Consumer <i>Smartphone, fitness watch</i>	~3 – 5 m (open sky); worse under canopy or near buildings	iPhone / Android, Garmin handheld	Navigation, rough site sketch
Mapping grade	~ 0.1 – 1 m (sub-meter) after correction	Trimble Geo / R1, Bad Elf, Emlid	Recording features, GIS data layers, shovel-test grids
Survey grade	~.5 – 2 cm horizontal (RTK / post-processed)	Trimble R-series + base station, total station	Precise excavation mapping, datums, point provenience

Note....

- Accuracy depends as much on technique (differential / RTK correction) as on the device.
- Vertical accuracy is always worse than horizontal — a phone good to ~3 m horizontally may be off 10+ m in elevation.

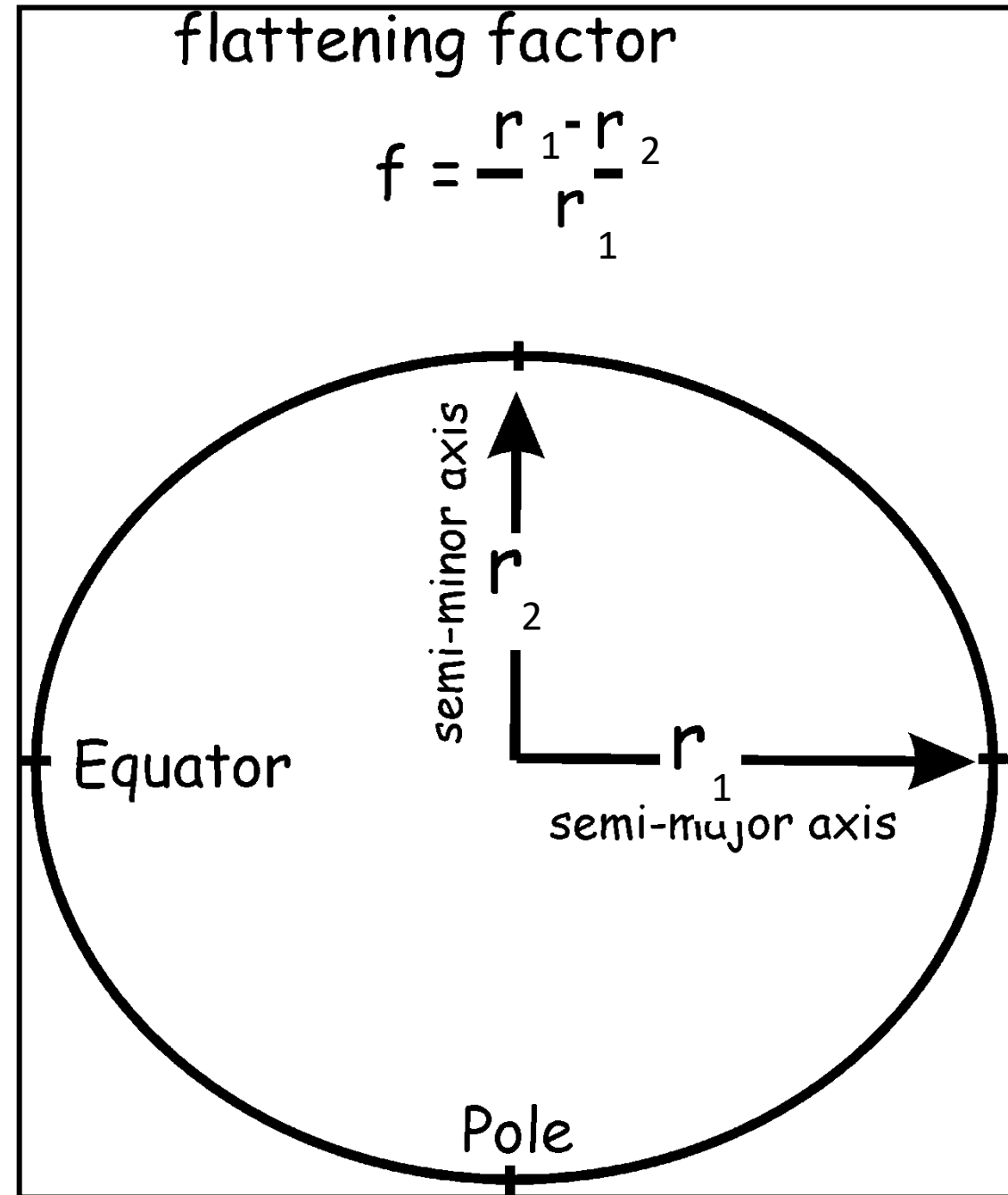
Datums, Projection, and Grids

Key Concepts

1. **Geodesy** –Science of measuring the shape of the Earth.

1. Geodesy

- Newton predicted that the earth should bulge at the equator because of rotation. Confirmed by French expeditions to Lapland and Peru in the 1730s.
- Ellipsoid size has been estimated from many places on earth
- But estimates differ across regions and continents.
- In 19th and early-20th centuries, different ellipsoids were developed for different parts of the world.
- Measurements were made using astronomical observations to establish latitudes and longitudes.
- Latitude/longitude are defined relative to a chosen ellipsoid, so the same physical spot gets different lat/long values depending on which ellipsoid is used.
- ***No ellipsoid model, no latitude or longitude***

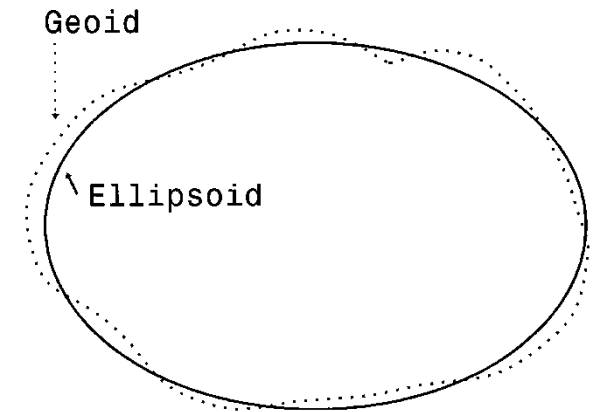
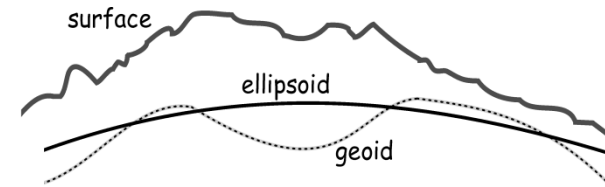


How to Estimate the Parameters for the Ellipsoid?

- The Earth's general shape is not uniform (duh!). But the ellipsoid is. How then to fit an ellipsoid model to locally variable topography?
- A step on the way to a solution: try to fit the ellipsoid to the **Geoid** (hat tip: Carl Friedrich Gauss).

Geoid:

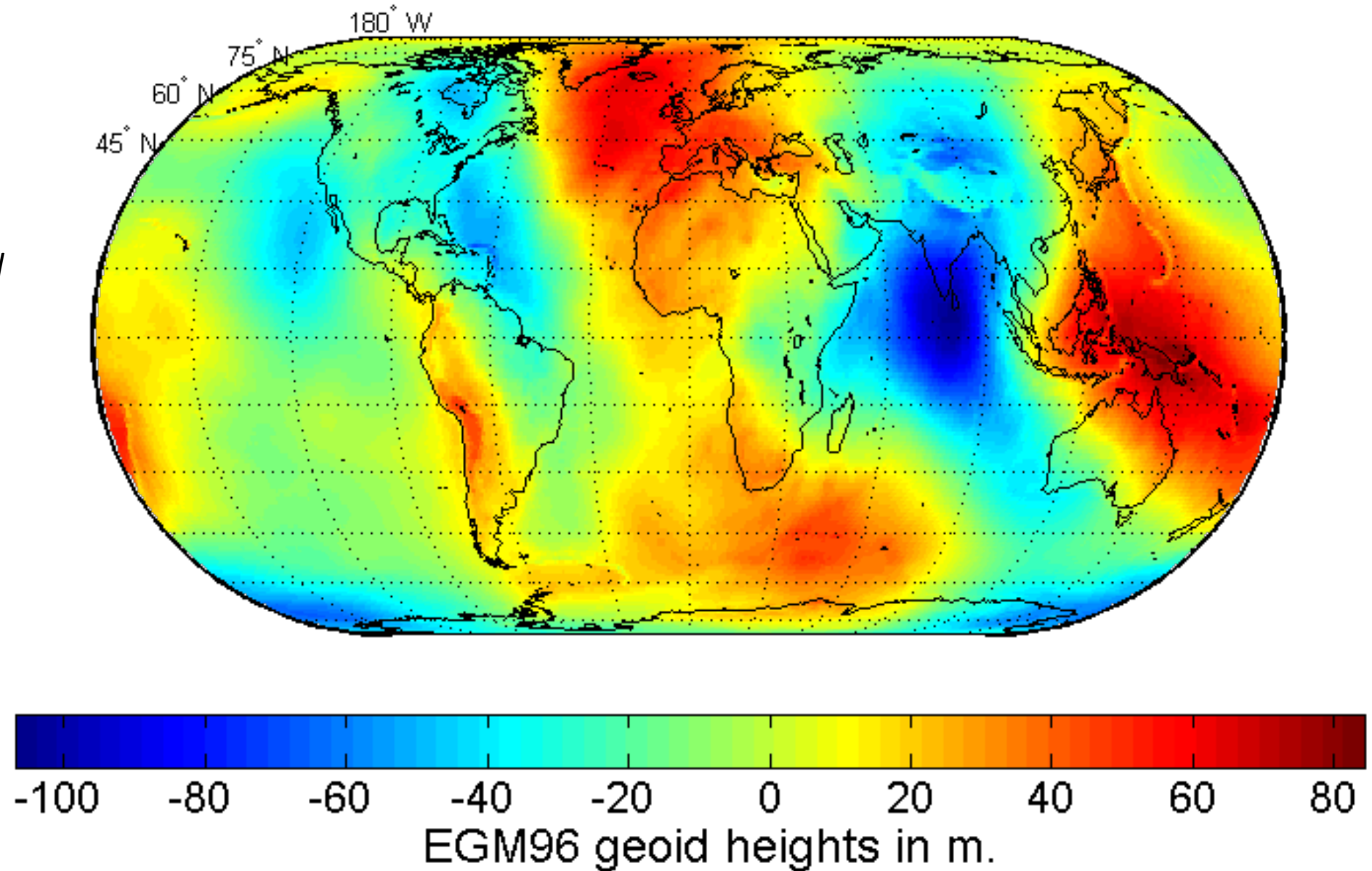
- gravity varies because mass is unevenly distributed
 - The **Geoid** surface can be thought of as the surface of a sea if it were to cover the entire earth.
 - This imaginary sea is not affected by the moon, wind, or waves – only gravity.
-
- Estimating the undulations in the earth's surface due to differential forces of gravity yields a **geoid model**.
 - An **ellipsoid model** is the smooth best-fit to that lumpy geoid model.
 - Early ellipsoid models defined to fit the LOCAL bumps of a geoid model.
 - With satellite measurements, modern ellipsoids fit the GLOBAL bumps.



Geoid Height Estimates

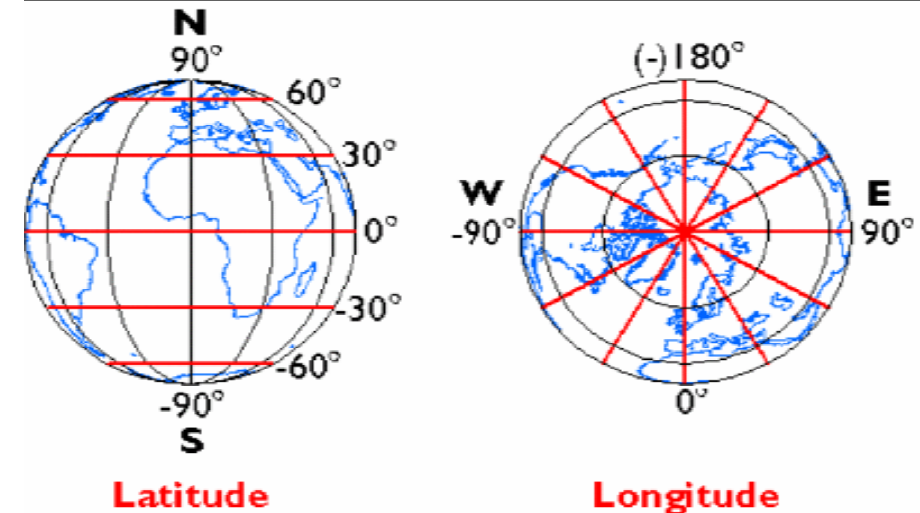
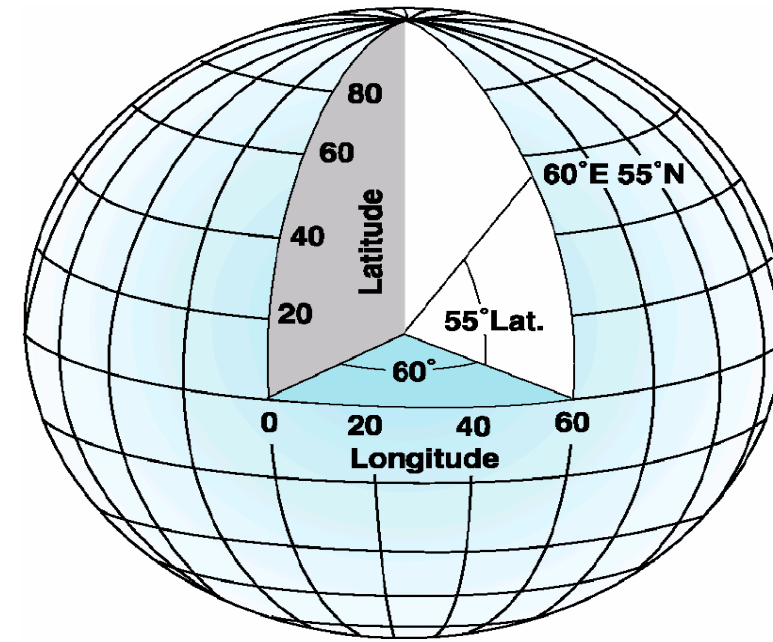
Relative to the WGS (World Geodetic System) 84 Ellipsoid Model

*Reds are where the geoid sits above the ellipsoid,
blues below;*



Latitude and Longitude

- An ellipsoid is defined by its major and minor axes.
- In turn, the poles and equator can be defined:
 - poles: the axis of revolution of the ellipsoid, around the minor axis
 - equator: circle mid-way between the poles spanning the widest part of the ellipsoid
- together these define **latitude**
- **Longitude** is defined arbitrarily:
 - 0 degrees = Greenwich Meridian
- ***Latitude/longitude are ANGLES measured from the Earth's center, not distances***



So how do we know where the longitudes and latitudes actually are on the ground?

We need a....

Geodetic Datum

Key Concepts

1. **Geodesy** – Science of measuring the shape of the Earth.
2. **Geodetic Datum** – Basis for defining accurate locations on the surface of the earth in the context of a reference ellipsoid.

2. Geodetic Datum

- A network of carefully controlled points on the ground, whose locations are given in terms of estimated ***latitude and longitude***.
- To create a **geodetic datum**:
 - specify an **ellipsoid model** of the earth
 - create a network of carefully surveyed locations, estimating lat/long for each
- Datums are updated as:
 - instruments get more accurate
 - ellipsoids get more accurate
- Older datums are based on **local ellipsoids**. In these cases, the initial reference point is located on the surface of the earth.
- Newer datums, are based on **geocentric ellipsoids** – the initial reference is the center of the earth
- .
- ***Lat/long coordinates measured in one geodetic datum do not match coordinates measured in a different datum***

The main datum used for Jamaica is ...

JAD69: Jamaican Datum of 1969.

- The reference ellipsoid is Clarke's 1866 ellipsoid
 - OK for mapping Jamaica, but poor fit to ellipsoid elsewhere.
- 1:50,000 Jamaica Metric Topographic Maps

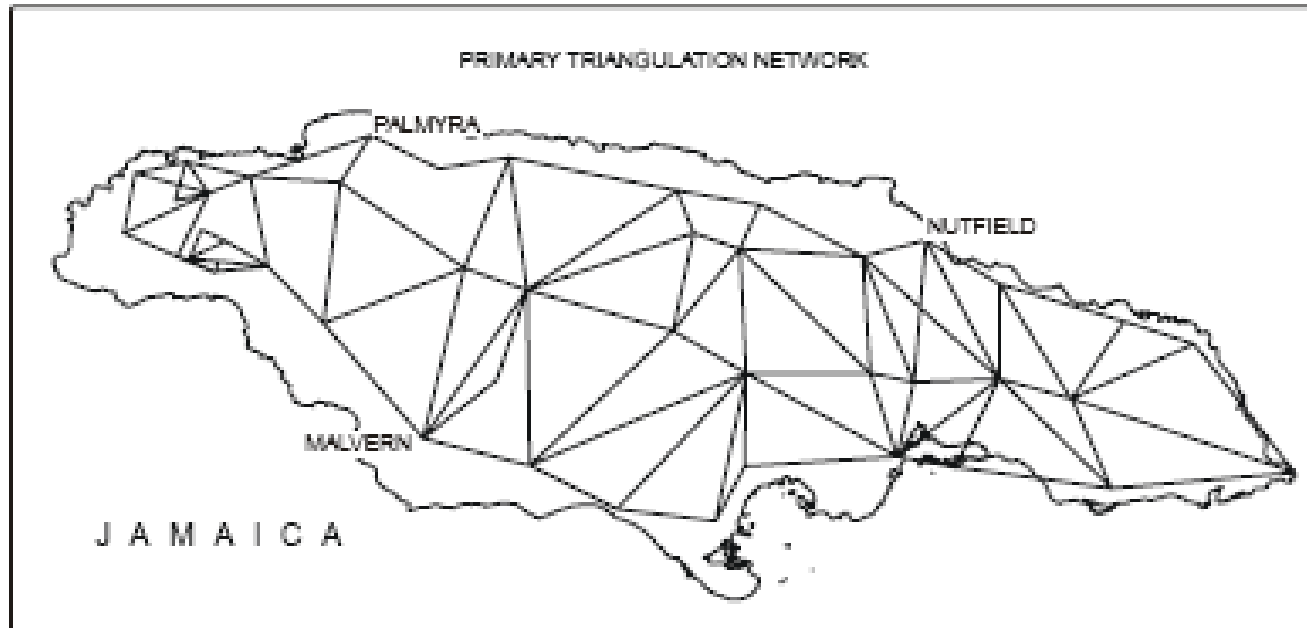


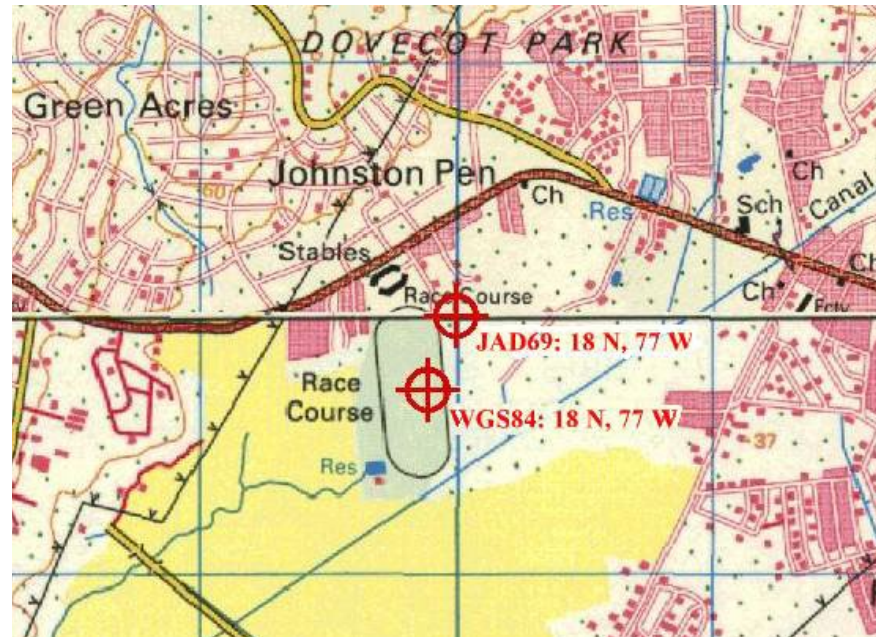
Figure 2. The Jamaica Primary Control Stations

WGS84: World Geodetic System of 1984.

- This datum was developed based on Doppler satellite measurements of the earth.
- The reference ellipsoid (with the same name (WGS84) is **geocentric**. So it is useful anywhere!

Warning!!

- Two points with the same Lat/Long in JAD69 and WGS84 can be up to **300 meters off!!**



Three main datums used for North America are:

NAD27, NAD83, and WGS84

- NAD27: North American Datum of 1927.

- The reference ellipsoid is Clarke's 1866 ellipsoid
- A local datum with its center in Kansas.

- NAD83: North American Datum of 1983.

- It included adjustments based on all survey points taken between the 20's and 80's.
- The reference ellipsoid used is GRS80, which is a **geocentric ellipsoid**.

- WGS84: World Geodetic System of 1984.

- This datum was developed based on Doppler satellite measurements of the earth.
- The reference ellipsoid (with the same name (WGS84) is **geocentric**,

- Coordinates measured in reference to the NAD27 Datum will be different than those measured in the more recent datums. The difference in positions can be off by up to 100 meters or more!

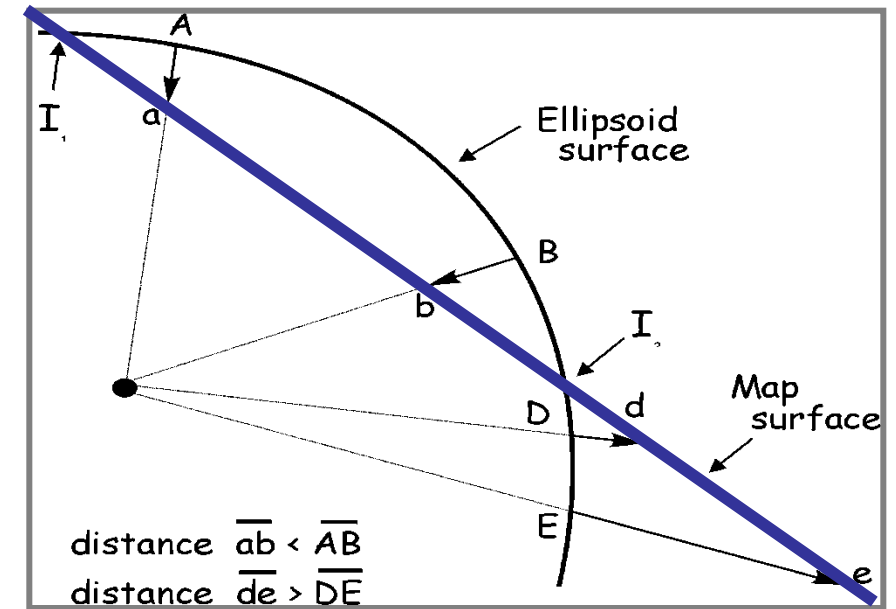
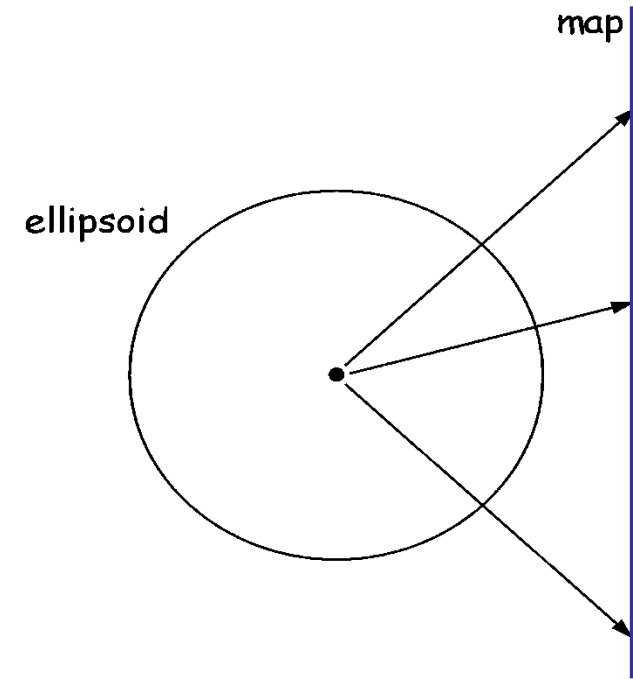
- GPS positions are determined to the WGS84 Datum as a default, although others can be specified.

Key Concepts

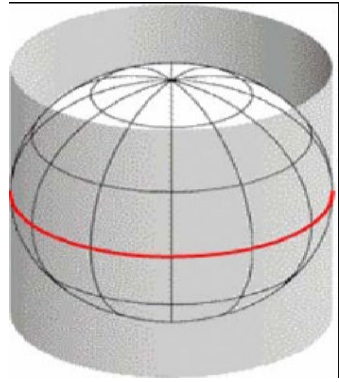
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3. **Map Projection** -the process of transferring points from the surface of the ellipsoid to a plane.

3. Map Projection

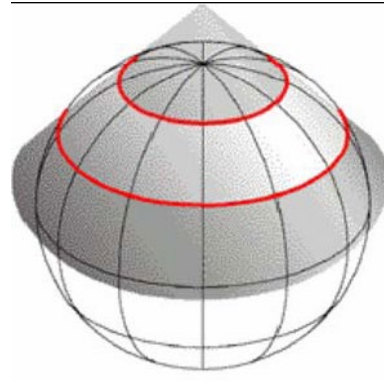
- The process of transferring points from the surface of the ellipsoid to a plane.
- ***All projections create distortion*** in distance, shape, area, or direction.
- But some are better than others.
- Which is better depends on shape what you are trying to map.



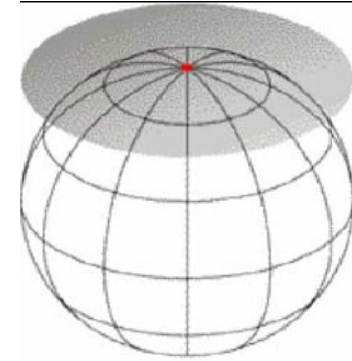
Three projection surfaces:



Cylindrical

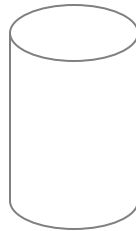


Conic



Planar (Azimuthal)

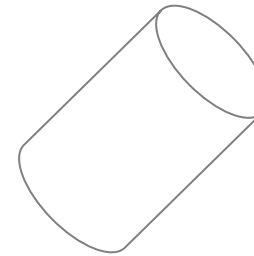
Three projection orientations:



Equatorial



Transverse



Oblique

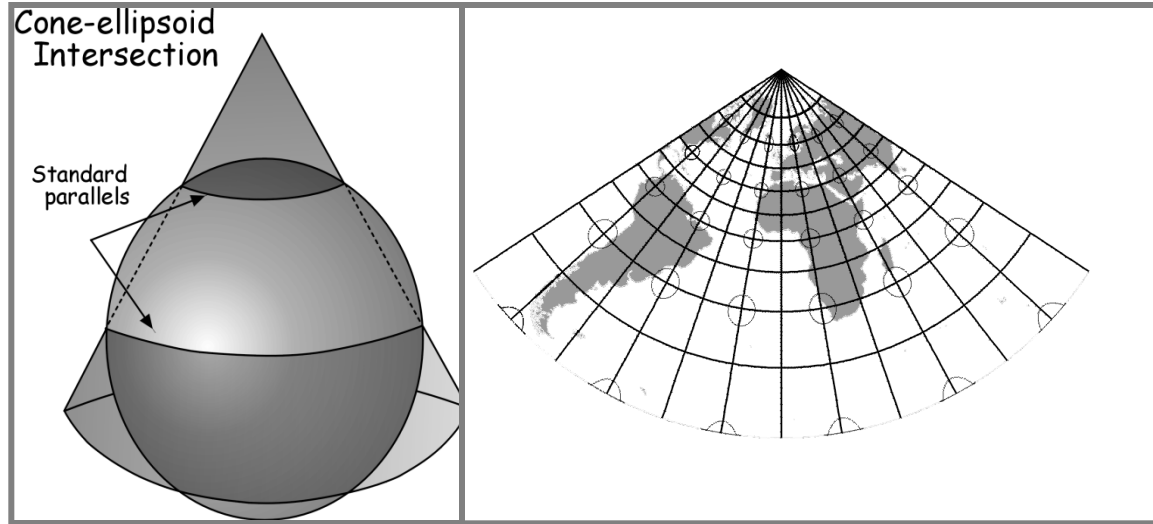
Two kinds of contact:

- tangent: being in contact at a single point or along a line
- secant: cutting, intersecting the surface:

Cooler Projections

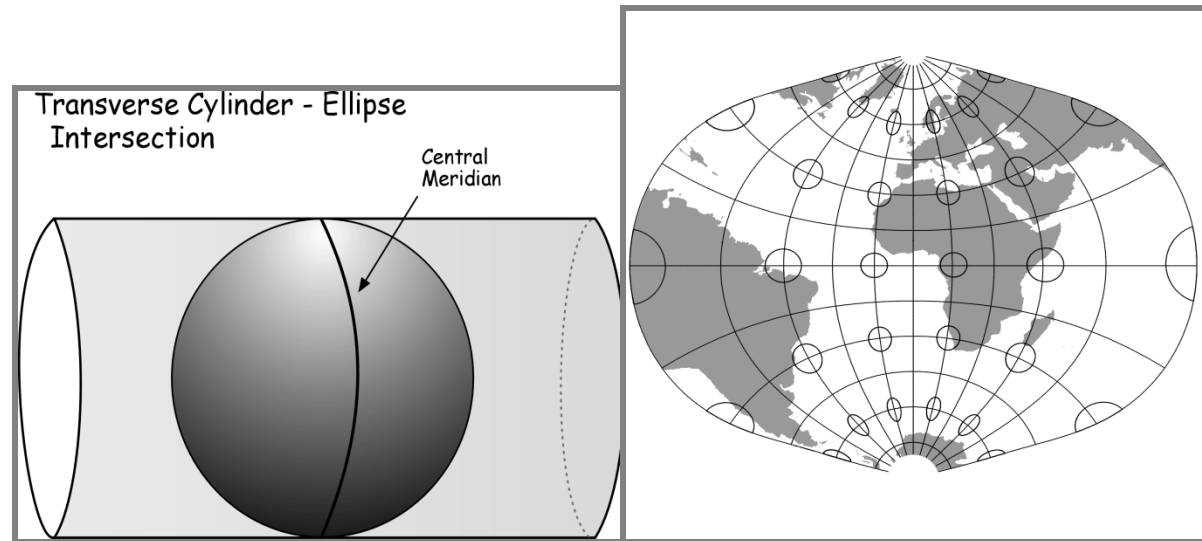
1. Lambert Conformal Conic:

A cone wrapped around the earth, such that the projection surface touches the ellipsoid surface along two standard parallels: Distortion is least along these E-W parallels



2. Transverse Mercator Projection:

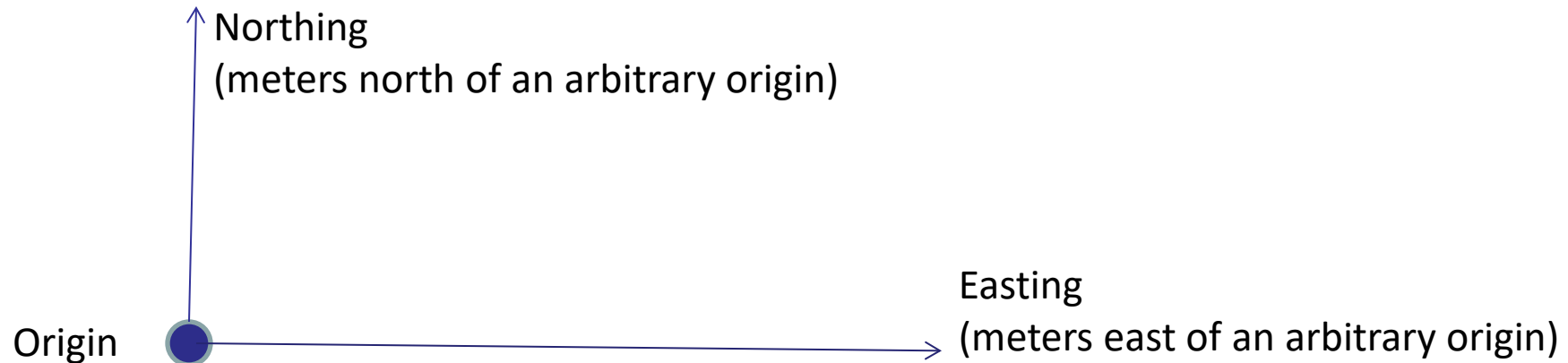
Transverse, cylindrical projection. The projection surface touches the ellipsoid surface along one central N-S meridian, and distortion is the least along this line.



Note: Both preserve local shape/angles, which is what you want for accurate mapping and survey work over limited areas.

Key Concepts

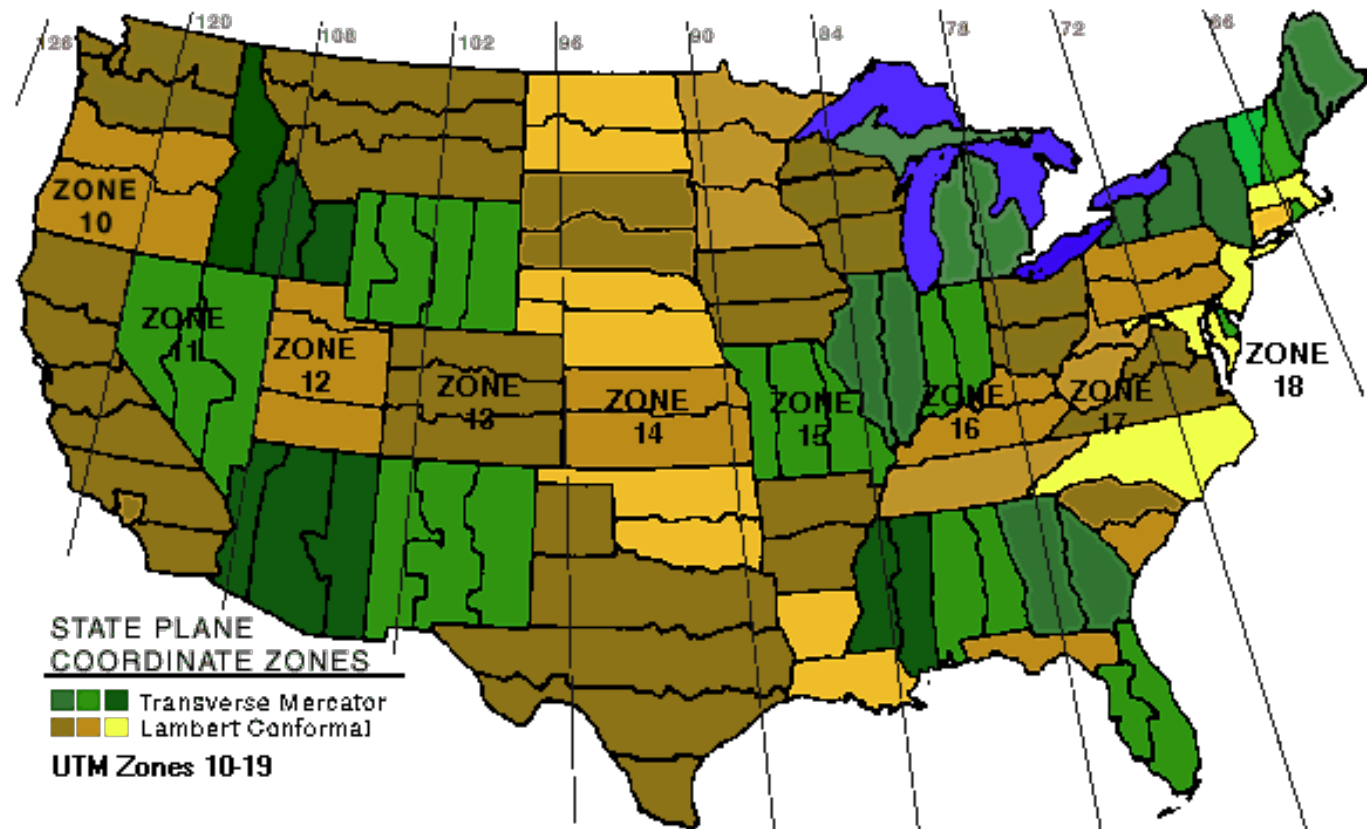
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4. **Coordinate Systems** – Allow us to use meters instead of degrees



4. Coordinate Systems

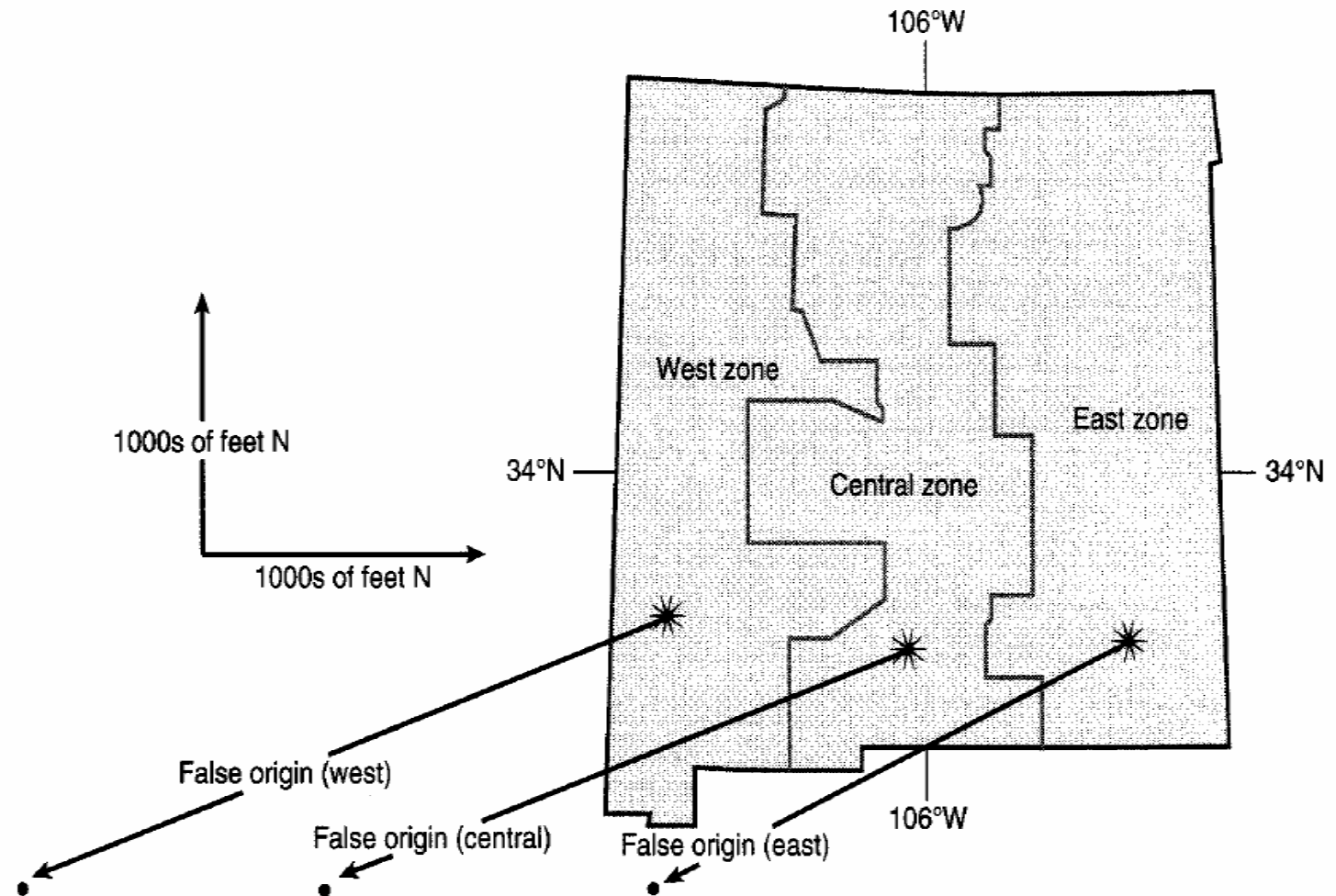
U.S. State Plane Coordinate System

- Not a projection
- Underlying projections:
 - Lambert Conformal Conic for “east-west” areas
 - Transverse Mercator for “north-south” areas



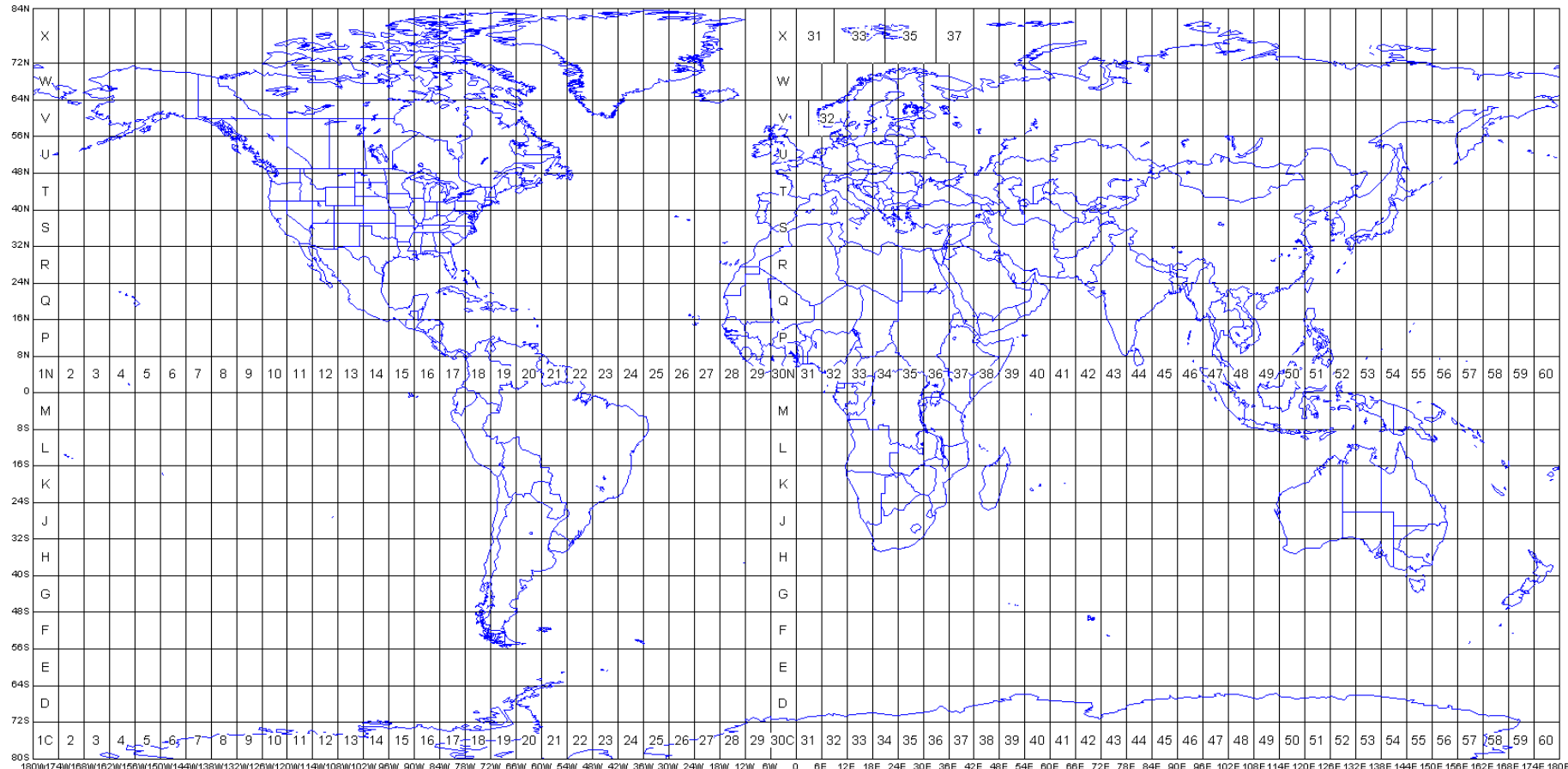
U.S. State Plane Coordinate System

- A “false” origin for each SP zone.
 - “false” northings and eastings for each SP zone.
-



Universal Transverse Mercator (UTM) Coordinate System

- Not a projection
- Underlying projection: Transverse Mercator
- **World** divided into 60 six-degree-wide zones
- Wide zones good for large study areas
- Coordinates discontinuous across zone boundaries



UTM Zone 11N

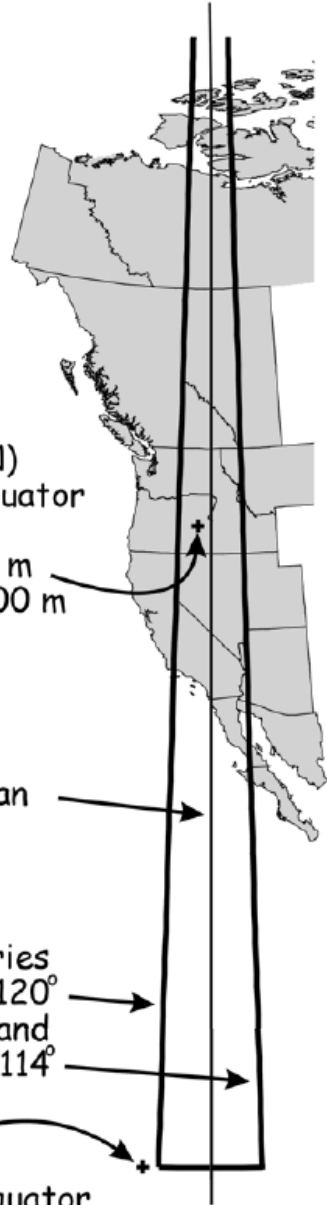
Coordinates are Eastings (E) relative to an origin 500,000 meters west of the zone central meridian, and a Northing (N) relative to the Equator

e.g., E = 397,800 m
N = 4,922,900 m

Central meridian
at W117°, zone
is 6° wide

Zone boundaries
at W120°
and
W114°

Origin
N = 0 at the Equator
E = 0 at 500,000 meters west of
the central meridian



“False” origin for each zone:

-To eliminate negative coordinate values to the west of the central meridian in each UTM zone, the origin of each is given a false easting value of 500,000 meters. This places the origin 500,000 meters west of the central meridian..

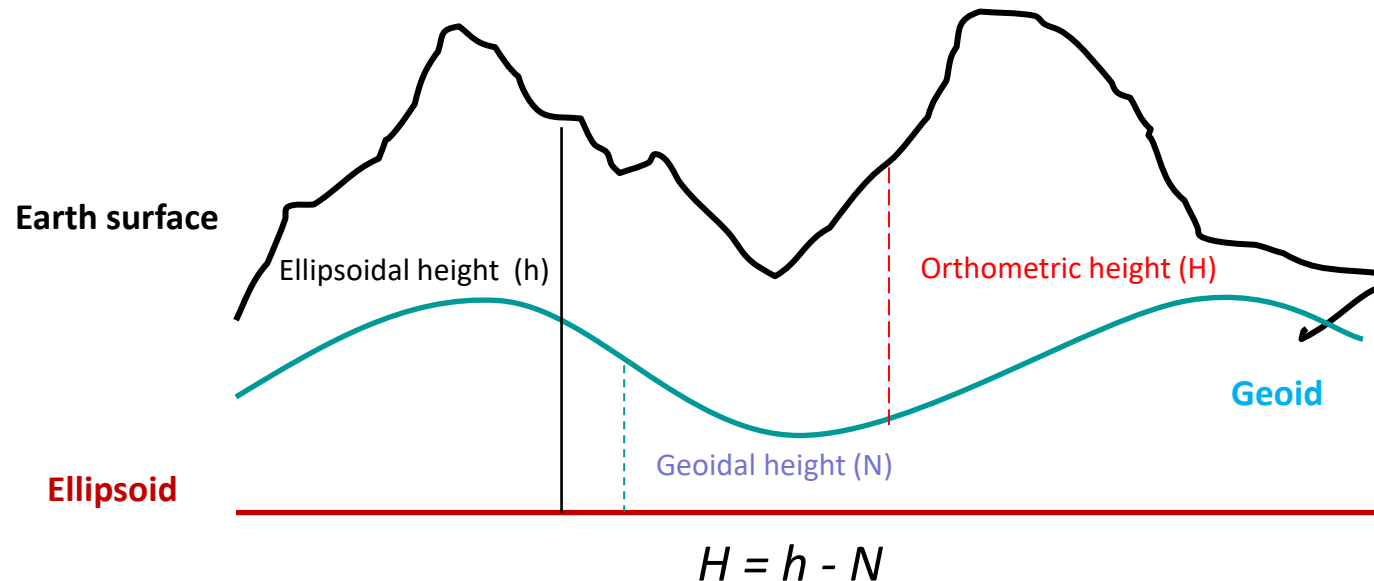
- The Equator is the origin for latitude, and it is zero for all Northern zones. Southern UTM zones have a false northing at the equator equal to 10,000,000 meters so that there are no negative latitudes for southern zones.

Key Concepts

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3. **Map projections** - The process of transferring points from the surface of the ellipsoid to a plane.
4. **Coordinate Systems** - Allow us to use meters instead of degrees
5. **Elevations** - Different kinds

5. Elevation

- Typically we think of measuring elevation with regard to the **mean sea level**. However, mean sea level is actually influenced by many factors such as tides and the moon's gravity, wind, and waves. So many "local" mean sea levels.
- For Globally comparable elevations, we need a better standard: ellipsoid or geoid model:
 - Ellipsoidal height: height of surface above ellipsoid
 - Orthometric height: height of surface above geoid.
 - Geoidal height: height of geoid above ellipsoid



Warning: GPS estimates of elevation are MUCH less accurate than Lat/Long or Northing/Easting coordinates.

Key Concepts

1. **Geodesy** – Science of measuring the shape of the Earth.
2. **Geodetic Datum** – Basis for defining accurate locations on the surface in the context of a reference ellipsoid.
3. **Map Projection** – Portrayal of the curved surface of Earth on a flat map
4. **Coordinate System** – Numeric representation of positions on a map projection: UTM, State Plane.
5. **Elevation** – three kinds!